HACK APERTUS 2026 · TRACK 2A · CHALLENGE BY OST

GeaNaVid

Every claim checked against the official voting booklet, with the line that decides it.

Gea · Na · Vid — Romansh for Yes · No · Blank

▶ Demo video | Built on Apertus v1.5 | Team Apertool · solo builder



THE PROBLEM

Voters hear the claim. The answer is on page 4 of 40.

THE CLAIM

„… die Bevölkerung seit Einführung der Personenfreizügigkeit im Jahr 2002 um rund 500'000 Personen gewachsen ist.“

“… the population has grown by about 500,000 since free movement began in 2002.”

OFFICIAL BOOKLET · PAGE 4

„Seit Einführung der Personenfreizügigkeit im Jahr 2002 ist die Bevölkerung um rund 1,7 Millionen Personen gewachsen.“

Na

CONTRADICTION

Claim–label pair from the OST demo set (MNLIoverSwissVotingBooklets, label 2). Booklet: Federal Council explanations for the vote of 14 June 2026, German edition, 40 pages, PDF page 4.

THE TASK

Booklet + claim in, a ballot mark out.

Voting booklet
PDF · German, French or Italian

Claim
one sentence · DE, FR or IT

→



GeaNaVid
CLI · Apertus v1.5

→

Gea	entailment	0
Na	contradiction	2
Vid	neutral	1

+ the passages it rests on · tokens used · seconds taken

JUDGED ON

Macro-F1 on a held-out benchmark · **evidence quality** · **input tokens** · **inference time**

All 9 claim/booklet language combinations. Labels 0/1/2 and judging criteria as set by OST in the Hack Apertus challenge brief.

THE NAME

Gea. Na. Vid.

Romansh, the fourth national language, for Yes, No and Blank. All three words come from the Romansh voting booklets.

Gea

Yes → `entailment`

„... recumondan cussegl federal e parlament da metter in **Gea** *en l'urna ...“*

“... the Federal Council and Parliament recommend putting a Yes in the ballot box ...”

`Booklet of 12 June 1994`

Na

No → `contradiction`

„Nus recumandain in **NA** *a las chapellinas blaus.“*

“We recommend a NO to the blue-helmet troops.” (referendum committee)

`Booklet of 12 June 1994`

Vid

Blank, empty → `neutral`

„... abitaziuns da vacanzas ch'èn savens **vidas** *...“*

“... holiday homes that mostly stand empty ...”

`Booklet of 25 June 1995`

Romansh text from the aligned corpus VotingBooklets-v1 (Stüssi & Vamvas, SwissText 2026, CC BY 4.0); English glosses ours.

THE CONTEXT

Narrow first, then let Apertus read.

1 **Top-5 passages of the right proposal**
the vote title narrows the booklet, BM25 picks the passages, the section the claim names first

≈ 1,300 tokens

CLAIM IN ANOTHER LANGUAGE? Apertus translates it into the booklet language first

2 **Translation**
one short call, then the same search

+ ≈ 300 tokens

full **The whole booklet**
the baseline OST recommends

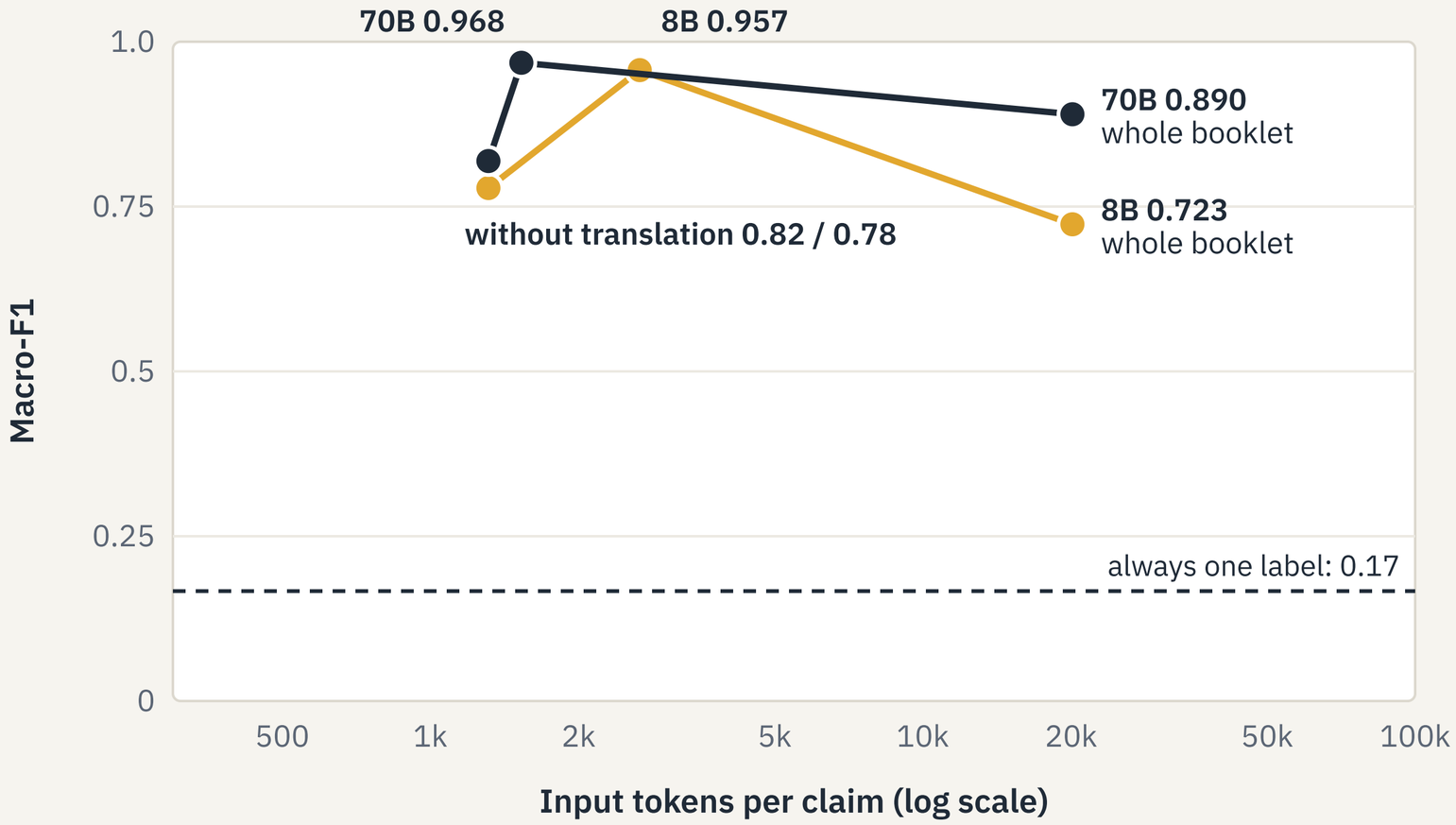
≈ 20,200 tokens

Per claim: 1,531 input tokens with 70B, 2,662 with 8B (three worked examples, a second look at neutral answers)

Mean input tokens per claim as reported by the API, OST v1.0 held-out test split (497 pairs), Apertus v1.5, 4 October 2026.

THE TRADE-OFF

Under a seventh of the tokens. Better verdicts.



Apertus 8B

GeaNaVid 0.957 with 2,662 tokens. Whole booklet 0.723 with 20,156.

Apertus 70B

GeaNaVid 0.968 with 1,531 tokens. Whole booklet 0.890.

Beginner task

Claim + reference text: 0.978 (8B) and 0.980 (70B).

OST MNLIoverSwissVotingBooklets v1.0, held-out test split: 497 pairs from 7 voting dates, settings chosen on development data. 95 % intervals about ±0.02. Unseen booklets (synthetic hold-out): 0.874 (8B) / 0.932 (70B) vs 0.665 / 0.872 whole booklet. 4 October 2026.

ACROSS LANGUAGES

Claim in Italian, booklet in German. The evidence stays German.

	DE	FR	IT
DE	0.98	0.94	0.91
FR	0.96	1.00	0.86
IT	1.00	0.96	0.98

Macro-F1 per pair, Apertus 8B. Rows: claim language · columns: booklet language. Framed: same language.

0.70 → 0.94

macro-F1 on the 331 cross-lingual test claims (8B), before and after Apertus translates the claim into the booklet language.

> The search runs on the translation, the evidence is quoted from the booklet as it is. No language field is needed: the claim language is detected.

OST v1.0 test split, 39 to 74 claims per pair, so single cells vary by about ±0.05. Same-language 0.987, cross-lingual 0.943 (8B); 70B 0.987 and 0.958.

EVIDENCE

Every verdict points to a line in the booklet.

Na `contradiction · label 2`

Claim (Italian): the reduced VAT rate falls from 2.5 to 2.4 per cent

> „Der dafür geltende reduzierte Mehrwertsteuersatz steigt von 2,5 auf 2,6 Prozent; …“

`25.09.2022 · DE · page 28 · AHV 21: additional financing by VAT`

`input tokens 2,794  time 0.8 s  Apertus 8B`

Quotes, not paraphrase

The evidence is the booklet sentence that decides, verbatim, with its PDF page, from the section the claim names, chosen after the verdict without another model call.

Evidence first, label second

Apertus names the passages before it gives the verdict, in strict JSON.

Not voting advice

GeaNaVid says what the booklet says. It never says how to vote.

OST v1.0 case v1-1484 (development split), live demo, 4 October 2026. On the test split the quoted sentence lies in OST's gold section for 86 % (8B) and 88 % (70B) of entailment/contradiction answers.

TEAM APERTOOL

Gea, Na or Vid.
Always with the line that decides it.

Niklas Wichter

Team Apertool · solo builder

REPO	github.com/NWichter/Hack_Apertus_2A_GeaNaVid
VIDEO	geanavid.apertool.ch/#video
REPORT	geanavid.apertool.ch/report.pdf
DOCS	technical documentation in the README

geanavid.apertool.ch

Hack Apertus 2026, Track 2A, challenge by OST. GeaNaVid is an independent entry, not a product of OST or the Federal Chancellery.